

TEST RIDE
A view of the Kavach interface inside a train engine

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1,465km

Or 2.1 per cent–the share of India's 68,000-km rail network that currently has Kavach

10

No. of collisions (out of 14) between 2018-19 and 2022-23 that were due to Signal Passed at Danger (SPAD)

in the locomotive. The train and the ground station are in constant communication with each other over radio frequencies (see *How Kavach Works*).

Simple though it may sound, Kavach is actually a complex web of devices installed at stations, on the track as well as in the engines. A radio-frequency identification device (RFID) on tracks carries location details. When a locomotive passes over it, the train picks up that information and relays it to the station. The system then continuously tells the locomotive its distance from signals, and at what speed it should approach or cross them. Any mismatch, and Kavach triggers an alarm. If the loco pilot fails to respond, the system applies brakes on its own.

As railways across the world began raising speeds, they realised the need for an ATP system. At higher speeds, it was felt, a driver may miss or misread signals on the tracks, leading to accidents. Automatic train protection was part of the European Train Control System (ETCS) that came into being in the '90s. India, though, needed its own technology for various reasons. Not only were the global versions on offer in Europe and the US expensive, but drivers jumping signals and accidents due to collisions were frightfully common in Indian Railways, and needed local solutions. Signal Passed at Danger (SPAD) is a grave offence, for which the erring driver may even lose his job. Between 2018-19 and 2022-23, out of the 14 cases of collisions reported, 10 have been due to SPAD. This, when the railways has been aiming to raise train speeds on key routes.

India faced a tricky situation when the first proof of concept for Kavach was produced in 2012. "When we started developing Kavach, India did not yet have 4G, and 2G was becoming obsolete for this technology application. We had to invent our own radio frequency," says Lalit Mansukhani of the Centre of Excellence at the Indian Railways Institute of Signalling and Telecom in Telangana. "We developed our own telecommunications backbone for railway use, which was compatible with the Long-Term Evolution